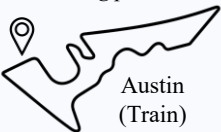


1 Racing Scenarios



80 starting positions



Austin
(Train)



Ego



Opp

left
center
right

720
scenarios
per track

3 opponent racelines
w/ 3 speed scales

slow

mid

fast

Noval Track Evaluation



Nürburgring



Moscow



Hockenheim

2 Expert Demonstration



Sense

2D LiDAR obstacle sensing



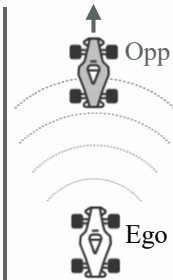
Plan

Frenet trajectory candidates



Select

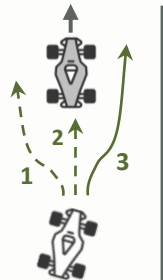
Balanced speed and clearance



Controller



pure pursuit



Action



expert
steer expert
speed

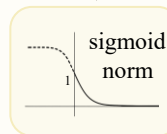
3 Imitation Learning



LiDAR scan



Ego speed



masked speed
embedding



concatenate

h_{t-1}

GRU

h_t

Action Head

predicted
steer

predicted
speed

Supervised Loss

$$\mathcal{L}_{\text{steer}} + 0.05 \mathcal{L}_{\text{speed}}$$



4 PPO Fine-tuning

Input Processing

GRU Backbone



Action Head

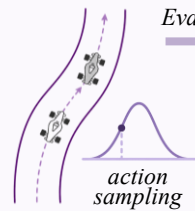


Value Head



Run

Update



Evaluate

progress
reward

collision
penalty

Learned Policy



Safety
>95%



Overtake
>90%